

Formulation of Research Question

A researcher must find a problem and formulate /define it for doing research.

The terms—**research topic**, **problem**, **question**, and **objective**—are closely related but serve distinct roles in shaping a research project.

1. Research Topic

- **What it is:** The general area or subject you want to explore.
- **Purpose:** Sets the broad context for your study.
- **Example:** *Digital transformation in retail HR.*

2. Research Problem

- **What it is:** A specific issue, gap, or challenge within the topic that needs investigation.
- **Purpose:** Justifies why the research is necessary.
- **Example:** *Retail HR departments struggle to adapt to digital tools, leading to inefficiencies.*

3. Research Question

- **What it is:** A precise question that your study aims to answer.
- **Purpose:** Guides the direction and scope of your research.
- **Example:** *How does digital transformation affect employee performance in retail HR?*

4. Research Objective

- **What it is:** A clear, actionable goal that outlines what you intend to achieve.
- **Purpose:** Breaks down the research question into measurable steps.
- **Example:**
 - *To assess employee perceptions of digital transformation.*
 - *To identify barriers to adoption of digital tools in retail HR.*

How They Connect

Element	Role in Research Flow
Topic	Identifies the area of research. Start Research
Problem	Identifies the issues / problem within the topic
Question	Frames the inquiry to address the problem
Objective	Defines the steps to answer the question

Research Question and Objectives:

Example 1: Environmental Science

Research Question:

How does microplastic pollution affect marine biodiversity in coastal ecosystems?

Research Objectives:

- To quantify the concentration of microplastics in selected coastal regions.
- To identify the species most affected by microplastic ingestion.
- To analyze changes in reproductive and feeding behavior due to microplastic exposure.

Example 2: Chemical Engineering

Research Question:

What is the efficiency of bio-based catalysts in converting waste cooking oil into biodiesel?

Research Objectives:

- To synthesize bio-based catalysts using agricultural waste.
- To compare the conversion rates of biodiesel using bio-based vs. conventional catalysts.
- To evaluate the environmental impact of the biodiesel production process.

Example 3: Mechanical Engineering

Research Question:

How can additive manufacturing improve the thermal performance of heat exchangers?

Research Objectives:

- To design and fabricate heat exchanger prototypes using 3D printing.
- To measure thermal conductivity and heat transfer rates of printed components.
- To assess the mechanical durability of printed heat exchangers under operational stress.

Example 4: Biomedical Engineering

Research Question:

Can wearable biosensors reliably detect early signs of dehydration in athletes?

Research Objectives:

- To develop a prototype wearable biosensor for hydration monitoring.
- To test sensor accuracy under various physical activity levels.
- To correlate biosensor data with clinical hydration markers.

Each objective breaks the question into measurable tasks, which helps to you build a structured and focused study.

WHAT IS A RESEARCH PROBLEM?

Some difficulty is being faced and solution is not well known.

A research problem does exist if the following conditions are met:

(i) Problem must be real. There must be an individual (or a group or an organisation), to whom the problem can be attributed. The individual or the organisation, occupies an environment, which is defined by values of the uncontrolled variables.

(ii) There must be at least two courses of action, to be pursued. A course of action is defined by one or more values of the controlled variables. For example,

Problem → heart disease → Blood pressure, Heart failure, stroke

Course of action → Decide what food to eat or not eat

→ Decide what exercise to do or not do

→ Decide what medicines to take

(iii) There must be at least two possible outcomes, of which one should be preferable to the other. In other words, this means that there must be at least one outcome that the researcher wants, i.e., an objective.

Blood pressure outcome: low, normal, high, very high

(iv) The courses of action available must provide some chance of obtaining the objective. But outcome of a course of action is not well known.

(v) There must be some environment(s) to which the difficulty pertains.

- there are several factors which may make the problem complicated.
- The environment may change affecting the efficiencies of the courses of action or the values of the outcomes;
- the number of alternative courses of action may be very large;
- persons not involved in making the decision may be affected by it and react to it favourably or unfavourably.

All such elements (or at least the important ones) may be thought of in context of a research problem.

SELECTING THE PROBLEM

Following points may be observed in selecting a research problem.

- Significant: Being faced in real life. Too narrow or too vague problems should be avoided.
- Novelty: Solution is not well known. Subject which is overdone should not be normally chosen.
- Feasible: To find solution with constraints. E.g. Controversial subject, unfamiliar area in which researcher has very little knowledge or expertise, time availability, costs, infrastructure, etc.

(The selection of a problem must be preceded by a preliminary study to ascertain its significance, novelty, and feasibility of finding a solution within available constraints.

DEFINING THE PROBLEM

The problem to be investigated must be defined unambiguously for that will help to discriminate relevant data from the irrelevant ones. A proper definition of research problem will enable the researcher to be on the track.

Questions like: What data are to be collected? What characteristics of data are relevant and need to be studied? What relations are to be explored. What techniques are to be used for the purpose?

Answers to all such questions can be determined only when the research problem has been well defined.

Steps to determine research question and objective

1. Explore and Identify a Broad Research Topic

- **Purpose:** Find an area that interests you and has academic or practical relevance.
- **How:**
 - Read recent articles, news, or journals.
 - Reflect on personal or professional experiences.
 - Consider societal or technological trends.
- **Example:** *Renewable energy technologies.*

2. Conduct a Literature Review

- **Purpose:** Understand what's already known and where gaps exist.
- **How:**
 - Search academic databases (Google Scholar, Scopus, etc.).
 - Identify recurring themes, debates, or unanswered questions.
- **Outcome:** Helps refine your topic and spot potential problems.

3. Define the Research Problem

- **Purpose:** Pinpoint a specific issue or gap within your topic.
- **How:**
 - Ask: *What's not working? What's missing? What needs improvement?*
 - Look for contradictions, inefficiencies, or unexplored areas.
- **Example:** *Solar panels lose efficiency in high humidity regions.*

4. Formulate the Research Question

- **Purpose:** Frame the problem as a question that guides investigation.
- **How:**
 - Use open-ended formats: *How, Why, What is the impact of...*
 - Ensure it's specific, measurable, and researchable.
- **Example:** *How does humidity affect the performance of solar panels in coastal regions?*

5. Set Research Objectives

- **Purpose:** Define what you aim to achieve—step-by-step goals.
- **How:**
 - Break the question into smaller, actionable tasks.
 - Use verbs like *analyze, evaluate, compare, design, measure.*
- **Example:**
 - *To measure solar panel efficiency under varying humidity levels.*
 - *To identify materials that perform better in humid conditions.*

Summary Table

Element	Description	Example
Topic	Broad area of interest	Renewable energy technologies
Problem	Specific issue or gap	Efficiency loss in humid climates
Question	Inquiry to address the problem	How does humidity affect solar panel output?
Objectives	Goals to answer the question	Measure, analyze, and propose solutions

Framing a Good Research Question

1. Clarity and Focus

- **Too broad:** *"How does technology affect education?"*
→ Narrow it: *"How does AI-based tutoring impact math performance in middle school students?"*
- **Too narrow:** *"How does one specific app affect one student's test score?"*
→ Broaden slightly to allow meaningful analysis.

2. Do a Deeper Literature Review

- Look for:
 - Gaps in existing research
 - Conflicting findings
 - Emerging trends
- Ask: *Has this question already been answered? If yes, can there be different approach having some advantage?*

3. Test It for Feasibility

Factors to be considered:

- Can the data needed be collect?
- Is the scope manageable within time and resources constraint?
- Is it measurable or answerable?

4. Get Feedback

- Talk to mentors, peers, or experts and people associated/ experiencing the problem.
- Ask: *Does this question make sense? Is it too vague or too complex?*

5. Refine the Language

- Use **precise terms**: Avoid jargon unless necessary.
- Make it **open-ended** (for qualitative) or **specific and measurable** (for quantitative).
- Example transformation:
 - Vague: *"Is solar energy good?"*
 - Refined: *"What are the economic and environmental impacts of solar energy adoption in rural India?"*

Common Pitfalls in Research Question Formulation

1. Being Too Broad or Too Vague

A question like *“How does technology affect society?”* is too sweeping to investigate meaningfully.

This can be narrowed down to a specific technology, population, and impact—e.g., *“How does smartphone use affect sleep quality among college students?”*

2. Being Too Narrow or Trivial

A question like *“What is the average number of pencils used by third-grade students at a particular school during the month of August?”* lacks broader relevance.

3. Lack of Feasibility

Asking something that requires data that can’t be accessed or methods can’t be used.

4. Ambiguous Language

- **Problem:** Vague terms like “better,” “impact,” or “effective” without context.
- **Fix:** Use precise, measurable language.

5. Ignoring the Literature

- **Problem:** Asking a question that’s already been answered or lacks grounding in existing research.
- **Fix:** Do a thorough literature review to identify gaps, contradictions, or emerging trends.

6. Multiple Questions in One

- **Problem:** Trying to answer too many things at once—e.g., *“How does diet and exercise and sleep affect mental health?”*
- **Fix:** Focus on one variable or break it into separate studies.

7. Lack of Alignment with Objectives

- **Problem:** Your question doesn’t match your research goals or methods.
- **Fix:** Make sure your objectives clearly support answering the question.

A good research question is:

- **Feasible**
- **Interesting**
- **Novel**
- **Ethical**
- **Relevant**